

Max Heap Definition

A **Max Heap** is a **complete binary tree** in which every parent node is **greater than or equal to its children**.

Therefore, the **largest element is always at the root**.

For an array representation:

- Parent of index i : $(i - 1) // 2$
- Left child: $2i + 1$
- Right child: $2i + 2$

Max Heap Algorithm

1. Build Max Heap

1. Start with the given array.
2. Find the last non-leaf node: $n // 2 - 1$.
3. Compare the node with its left and right children.
4. Find the largest among the parent and children.
5. If a child is larger, swap it with the parent.
6. Continue heapifying down the tree.
7. Repeat for all non-leaf nodes.
8. The resulting array is a Max Heap.

2. Heap Sort Using Max Heap

1. Build a Max Heap from the array.
2. The largest element will be at the root.
3. Swap the root with the last element.
4. Reduce the heap size by one.
5. Heapify the root again.
6. Repeat until all elements are sorted.

Output:

```
Enter elements separated by spaces: 2
Original array: [2]
Sorted array: [2]

...Program finished with exit code 0
Press ENTER to exit console.[]
```

Time Complexity

Operation	Time Complexity
Build Max Heap	$O(n)$
Insert element	$O(\log n)$
Delete maximum	$O(\log n)$
Heapify	$O(\log n)$
Heap Sort	$O(n \log n)$